

Probability Rules Sets 1–4 · sample spaces, events, set operations	
Look for: "find P(A)", "what is the probability that...", "P(A and B)", "P(A or B)", "A or B but not both", "mutually exclusive", "complement"	
Complement: $P(A') = 1 - P(A)$	Independent $(A \perp B)$: $P(A \cap B) = P(A) \cdot P(B)$ and $P(A B) = P(A)$
Addition: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$	<i>"P(defective) = 0.03, P(defective AND returned) = 0.01 → P(returned defective) = 0.01/0.03 = 0.333"</i> <i>"A card is drawn. A=heart, B=face card. P(A∩B)=3/52. P(A B)=3/12=0.25=P(A)=13/52=0.25 → independent"</i>
Mutually exclusive $(A \cap B = \emptyset)$: $P(A \cup B) = P(A) + P(B)$	
Multiplication: $P(A \cap B) = P(A B) \cdot P(B) = P(B A) \cdot P(A)$	

Distribution	Conditions / When to use	PMF / PDF & Key Formulas	Mean & Variance	Standardize / Notes
Discrete RV <i>Sets 8–10</i>	Table of x values & probabilities $f(x)$; "find E(X)", "find V(X)", CDF	$F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$ $E(X^2) = \sum x^2 f(x)$ <i>Linear:</i> $E(aX + b) = a\mu + b \cdot V(aX + b) = a^2\sigma^2$	$\mu = \sum x f(x)$ $\sigma^2 = E(X^2) - \mu^2$	<i>No standardization — use table directly</i>
$X \sim B(n, p)$ <i>Binomial · Set 11</i>	n fixed trials; P(success) = p ; independent; X = # successes; two outcomes	$b(x; n, p) = \binom{n}{x} p^x (1 - p)^{n-x}$	$\mu = np$ $\sigma^2 = npq$ $q = 1 - p$	If $np \geq 5$ & $nq \geq 5$: $Z = \frac{x - np}{\sqrt{npq}} \sim N(0, 1)$ <i>Poisson approx: if $n \geq 100$, $np \leq 10 \rightarrow \lambda = np$</i>
$X \sim \text{Pois}(\lambda)$ <i>Poisson · Set 12</i>	Events per time/area; avg rate λ ; rare events; no fixed n	$p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots$ <i>Scale: rate r/unit over t units $\rightarrow \lambda = r \cdot t$</i>	$\mu = \lambda$ $\sigma^2 = \lambda$	<i>"5 calls/hr, 2 hrs" $\rightarrow \lambda = 10$</i> <i>"2 errors/page, next page" $\rightarrow e^{-2}$</i>
$X \sim U(A, B)$ <i>Uniform · Sets 13–14</i>	Equally likely over interval $[A, B]$	$f(x) = \frac{1}{B - A}, A \leq x \leq B$ $P(a \leq X \leq b) = \frac{b - a}{B - A}$ $F(x) = \frac{x - A}{B - A}, \text{median: } F(m) = 0.5$	$\mu = \frac{A + B}{2}$ $\sigma^2 = \frac{(B - A)^2}{12}$	<i>"U(7,10). P(≤ 8.8)?" $\rightarrow (8.8 - 7)/3 = 0.60$</i>
$X \sim N(\mu, \sigma^2)$ <i>Normal · Sets 15–17</i>	Bell-shaped, symmetric; mean=median=mode; use z-table for probabilities	$P(a < X < b) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right)$ $P(Z > z) = 1 - \Phi(z) \cdot P(Z < -z) = 1 - \Phi(z)$ <i>Percentile: look up z in table body $\rightarrow x = \mu + z\sigma$</i>	$E(X) = \mu$ $V(X) = \sigma^2$	$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$ <i>Table gives $\Phi(z) = P(Z \leq z)$ (left-tail area)</i>
$X \sim \text{Exp}(\lambda)$ <i>Exponential · Set 18</i>	Time between events / waiting time / component life; memoryless property	$f(x) = \lambda e^{-\lambda x}, x \geq 0$ $P(X \leq x) = 1 - e^{-\lambda x} \cdot P(X > x) = e^{-\lambda x}$	$\mu = 1/\lambda$ $\sigma^2 = 1/\lambda^2$	Memoryless: $P(X > s + t \mid X > s) = P(X > t)$ <i>"Mean life=15d" $\rightarrow \lambda = 1/15$</i>
Sample Mean $\bar{X}$ <i>CLT · Sets 19–20</i>	i.i.d. sample size n ; normal pop $\rightarrow$ any n ; any pop $\rightarrow n \geq 30$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ <i>Sample size for CI: $n = \left(\frac{z_{\alpha/2} \sigma}{E}\right)^2$</i>	$E(\bar{X}) = \mu$ $\text{SE}(\bar{X}) = \sigma/\sqrt{n}$ $V(\bar{X}) = \sigma^2/n$	$\bar{X} \sim N(\mu, \sigma^2/n)$ <i>CLT applies when $n \geq 30$</i>
Sample Prop. $\hat{p}$ <i>CLT · Set 21</i>	$np \geq 5$ & $nq \geq 5$ required for normal approx; $\hat{p} = x/n$	$Z = \frac{\hat{p} - p}{\sqrt{pq/n}} \sim N(0, 1)$ <i>Sample size: $n = z_{\alpha/2}^2/(4E^2)$ if p unknown</i>	$E(\hat{p}) = p$ $\text{SE}(\hat{p}) = \sqrt{pq/n}$ $V(\hat{p}) = pq/n$	$\hat{p} \sim N(p, pq/n)$ $q = 1 - p$

Single Mean Set 22 · one sample, testing/estimating μ
Look for: "mean/average of one group", "Is μ greater/less than X ?", "estimate the mean", "one sample"
Case I — σ known (any n) $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \quad \cdot \quad \text{Cl: } \bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$
Case II — σ unknown, $n \geq 30$ $Z = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim N(0, 1) \quad \cdot \quad \text{Cl: } \bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$
Case III — σ unknown, $n < 30$, normal pop. $T = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1} \quad \cdot \quad \text{df} = n - 1 \quad \cdot \quad \text{Cl: } \bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}$

Single Proportion Set 23 · $\hat{p} = x/n$
Look for: "proportion/% who...", yes/no outcome, one group, "Is the rate greater than p_0 ?" <i>"200 voters; 54% support bill. Evidence >50%?" $\rightarrow p_0=0.5$, one-tail "80 items, 12 defective. 90% CI for defect rate." $\rightarrow$ CI, use $\hat{p}$</i>
$Z = \frac{\hat{p} - p_0}{\sqrt{p_0 q_0/n}} \sim N(0, 1) \quad \text{test} \rightarrow \text{use } p_0$ Cl: $\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p} \hat{q}}{n}} \quad \text{Cl} \rightarrow \text{use } \hat{p}$ $n = \frac{z_{\alpha/2}^2 \hat{p} \hat{q}}{E^2} \quad [\hat{p} = 0.5 \text{ if unknown; } n = z^2/(4E^2)]$

Conditional Probability & Bayes' Theorem Sets 5–7
Look for: "given that...", "knowing that...", "P(B A)", "prior probability", "what fraction of defectives came from machine X ?", tree diagram problems
$P(A B) = \frac{P(A \cap B)}{P(B)}$ Conditional probability
Total probability: $P(A) = \sum_i P(A B_i) P(B_i)$ partition $B_1, \dots, B_k$
Bayes: $P(B_k A) = \frac{P(A B_k) P(B_k)}{\sum_i P(A B_i) P(B_i)}$ <i>"3 machines B_1, B_2, B_3 make 30%/45%/25% of product; defect rates 2%/3%/2%. Given defective, P(B_i)?" $\rightarrow$ Bayes</i>

Two Independent Means Sets 24–25 · two separate groups, compare μ_1 and μ_2
Look for: "compare means of two separate/independent groups", "do A and B differ in average?" — not the same people measured twice (that's paired)
Choose case: σ_1, σ_2 given $\rightarrow$ I $n_1 \geq 30$ & $n_2 \geq 30 \rightarrow$ II small n , $\max(s)/\min(s) \leq 1.4 \rightarrow$ III (pooled) ratio $> 1.4 \rightarrow$ IV (unpooled)
Case I — σ_1, σ_2 known (any n) $Z = \frac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \sim N(0, 1) \quad \cdot \quad \text{Cl: } (\bar{x} - \bar{y}) \pm z_{\alpha/2} \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$
Case II — σ unknown, $n_1 \geq 30$ & $n_2 \geq 30$ $Z = \frac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} \sim N(0, 1) \quad \cdot \quad \text{Cl: } (\bar{x} - \bar{y}) \pm z_{\alpha/2} \sqrt{s_1^2/n_1 + s_2^2/n_2}$
Case III — small n, ratio ≤ 1.4 (Pooled T) $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \quad \cdot \quad \text{df} = n_1 + n_2 - 2$ $T = \frac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0}{\sqrt{s_p^2(1/n_1 + 1/n_2)}} \sim t_{n_1 + n_2 - 2}$ Cl: $(\bar{x} - \bar{y}) \pm t_{\alpha/2, \nu} \sqrt{s_p^2(1/n_1 + 1/n_2)}$
Case IV — small n, ratio > 1.4 (Unpooled/Welch T) $T = \frac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} \sim t_\nu$ $\nu = \left\lfloor \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \right\rfloor$ Cl: $(\bar{x} - \bar{y}) \pm t_{\alpha/2, \nu} \sqrt{s_1^2/n_1 + s_2^2/n_2}$

Regression & Correlation			
Sets 28–30 · $\hat{y} = b_0 + b_1 x$ · always compute s_{xy}, s_x^2, s_y^2 first			
Look for: "predict Y from X", "linear relationship", "for each unit increase in X, Y changes by...", "correlation", "fit a line to the data"			
Variables: x = predictor (independent/explanatory) · y = response (dependent/outcome) · $\hat{y}$ = predicted value of y · $\mathbf{b}_0$ = y-intercept (value of y when $x=0$) · $\mathbf{b}_1$ = slope (change in y for each +1 unit of x) · $\boldsymbol{\beta}_1$ = true (population) slope · $\mathbf{e} = y - \hat{y}$ = residual (actual minus predicted, positive = above line)			
$\mathbf{s}_{xy}$ = sample covariance of x & y · $\mathbf{s}_x^2, \mathbf{s}_y^2$ = sample variances of x and y · r = sample correlation coefficient ($-1 \leq r \leq 1$; sign = direction, magnitude = strength) · ρ = true (population) correlation · $\mathbf{R}^2$ = coefficient of determination = % of variation in y explained by x · SSE = sum of squared residuals · $\mathbf{s}_e$ = standard error of regression (typical distance of points from the line)			
Step 1 — Compute summary statistics		Step 2 — Regression line	
$s_{xy} = \frac{\sum x_i y_i - (\sum x_i)(\sum y_i)}{n}$		$b_1 = s_{xy}/s_x^2 \cdot b_0 = \bar{y} - b_1 \bar{x}$	
$s_x^2 = \frac{\sum x_i^2 - (\sum x_i)^2/n}{n-1}$ (same form for s_y^2)		$\hat{y} = b_0 + b_1 x \cdot \text{residual: } e = y - \hat{y}$	
Compute $\sum x_i y_i$ by hand (multiply pairs, sum up)		Step 3 — Correlation & fit	
		$r = s_{xy}/(s_x \cdot s_y) \cdot R^2 = r^2$	
		$\text{SSE} = (n-1)(s_y^2 - s_{xy}^2/s_x^2) \cdot s_e = \sqrt{\text{SSE}/(n-2)}$	
		Step 4 — Hypothesis tests (both use t_{n-2})	
		$s_{b_1} = \frac{s_e}{\sqrt{(n-1)s_x^2}}$ e.s.e. of slope b_1	
		Test slope: $T = b_1/s_{b_1} \sim t_{n-2} \cdot H_0: \beta_1 = 0$ (no relationship)	
		Test corr: $T = r\sqrt{(n-2)/(1-r^2)} \sim t_{n-2} \cdot H_0: \rho = 0$	
		Both tests are equivalent — same t_{obs} and same p-value.	
z critical values			
Two-tail: 90%→1.645 · 95%→1.96 · 99%→2.576			
One-tail: 90%→1.28 · 95%→1.645 · 99%→2.33			
Other: 94%→1.88 · 98%→2.326 · 99.5%→2.807			
p-value from z-table (table gives $P(Z < z)$):			
$H_1: \theta > \text{val} \rightarrow p = 1 - \Phi(z_{\text{obs}})$ upper tail			
$H_1: \theta < \text{val} \rightarrow p = \Phi(z_{\text{obs}})$ lower tail			
$H_1: \theta \neq \text{val} \rightarrow p = 2(1 - \Phi(z_{\text{obs}}))$ two-tail			
Bracket p-value from t-table (same tail rules):			
$t_{\text{obs}} = 2.49$, df= 11, one-tail:			
$t_{0.025, 11} = 2.201 < 2.49 < t_{0.01, 11} = 2.718$			
$\rightarrow 0.01 < p < 0.025$			
Two-tail: double the bracket.			
Key traps:			
Prop CI $\rightarrow$ use $\hat{p}$; test $\rightarrow$ use p_0			
2-prop test $\rightarrow$ pool $\hat{p}_p$; CI $\rightarrow$ no pool			
$(\mu_1 - \mu_2)_0 = 0$ unless claim says "by k "			
Before/after same subject $\rightarrow$ paired, not 2-sample			
χ^2 always upper tail only			